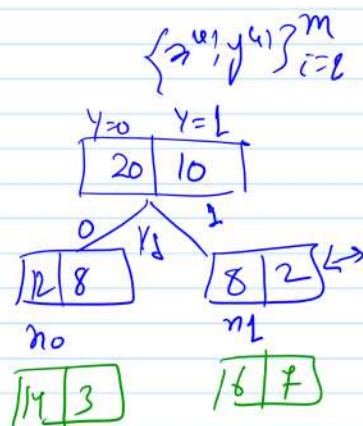
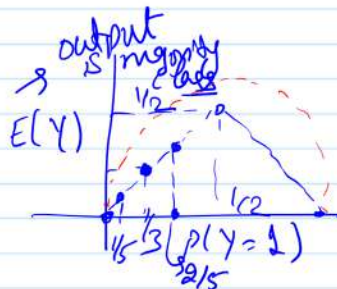




COL 774
Oct 8, 2025

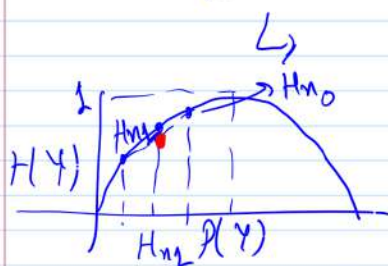


$$\begin{aligned}
 E_n &= 19/30 = 1/3 \\
 E_{n_0} &= 8/20 = 2/5 \\
 E_{n_1} &= 2/10 = 1/5 \\
 E_{split} &= p_{n_0} * E_{n_0} + p_{n_1} * E_{n_1} \\
 &= \frac{2}{3} * \frac{2}{5} + \frac{1}{3} * \frac{1}{5} \\
 &= \frac{4}{15} + \frac{1}{15} = \frac{5}{15} = \frac{1}{3}
 \end{aligned}$$

Entropy:- (Measures the amount of randomness)
 is in a distribution)
 uncertainty

y :- Random variable $\{y_1, \dots, y_r\}$
 $-\{ \log p(y=y_k) \}$:- surprise associated with observing y_k as the random value

$H(Y)$ = Expected surprise associated with R.V. y



$$\sum_{y_k} p(y=y_k) [-\log p(y=y_k)]$$

$$\begin{aligned}
 Y \in \{0, 1\} \\
 p(y=1) &= 1/2 \\
 H(Y) &= \frac{1}{2} * 1 + \frac{1}{2} * 1 \\
 &= 1
 \end{aligned}$$

$$(At no) = -H(Y) = -\left[\frac{1}{3} \log \frac{1}{3} + \frac{2}{3} \log \frac{2}{3} \right]$$

Conditional Entropy:- $H(Y|X_2) = \sum_{y_j} p(X_2=y_j) H(Y|X_2=y_j)$

$$H(Y|X_2=y_j) = \sum_{y_k} -p(y=y_k | X_2=y_j) \log p(y=y_k | X_2=y_j)$$

choose x_j :- s.t
 $I(Y, X_2) = [H(Y) - H(Y|X_2)]$ is maximised
 Mutual Information between Y & X_2

$$I(Y, X_2) = H(Y) - H(Y|X_2)$$

(in our example)

$$p(X_2=0) H(Y|X_2=0) + p(X_2=1) H(Y|X_2=1)$$

$$H(Y|X_0=0) = -\left[\frac{3}{5} \log \frac{3}{5} + \frac{2}{5} \log \frac{2}{5} \right]$$

$$H(Y|X_0=1) = -\left[\frac{4}{5} \log \frac{4}{5} + \frac{1}{5} \log \frac{1}{5} \right]$$

$$P(X_0=0) = \frac{2}{3} \quad P(X_0=1) = \frac{1}{3}$$